

TECHNICAL DOCUMENTATION

RAG-BASED CUSTOMER SUPPORT ASSISTANT

1. INTRODUCTION

This project implements a Retrieval-Augmented Generation (RAG) system for customer support automation. It enables users to query PDF-based knowledge sources and receive intelligent, context-aware responses.

The system combines:

- Document retrieval
- Embedding-based search
- Language model generation
- Workflow orchestration
- Human fallback system

2. SYSTEM OVERVIEW

The system is designed to replace traditional keyword-based search systems with a semantic AI-powered assistant.

It works by:

- Understanding user queries
- Retrieving relevant document sections
- Generating responses using context
- Ensuring accuracy through confidence scoring

3. DESIGN DECISIONS

Chunking Strategy:

Small overlapping chunks were used to maintain context continuity and improve retrieval quality.

Embedding Model:

Sentence Transformers were chosen for their balance between accuracy and performance.

Vector Database:

ChromaDB was used due to its simplicity, speed, and support for similarity search.

Workflow Engine:

LangGraph was used to structure execution flow and decision-making.

4. SYSTEM WORKFLOW

The system follows this pipeline:

1. User enters query
2. Query is converted into embeddings
3. Similar chunks are retrieved from vector DB
4. Context is passed to generator
5. Response is created
6. Confidence is calculated
7. HITL triggered if required
8. Final output returned

5. KEY CONCEPTS USED

RAG (Retrieval Augmented Generation)

Combines retrieval + generation to improve accuracy.

Embeddings

Convert text into numerical vectors for semantic understanding.

Vector Search

Finds semantically similar content instead of keyword matching.

HITL

Human fallback system for handling uncertain responses.

6. CHALLENGES

- Determining optimal chunk size
- Balancing retrieval accuracy
- Designing confidence scoring logic

- Managing workflow transitions in LangGraph

7. LIMITATIONS

- Depends on quality of input PDF
- Retrieval may fail for ambiguous queries
- No memory-based conversation tracking
- Rule-based confidence scoring is simple

8. FUTURE ENHANCEMENTS

- Multi-document support
- Advanced LLM integration (GPT/Gemini)
- Chat memory system
- UI-based interface (Streamlit/Web app)
- Hybrid search (BM25 + embeddings)

9. CONCLUSION

The RAG-Based Customer Support Assistant demonstrates a complete AI pipeline combining retrieval systems, language models, and workflow orchestration.

It improves traditional customer support systems by:

- Providing accurate contextual answers
- Reducing manual workload
- Ensuring reliability through HITL
- Structuring execution using LangGraph

This project represents a practical implementation of modern AI system design principles.